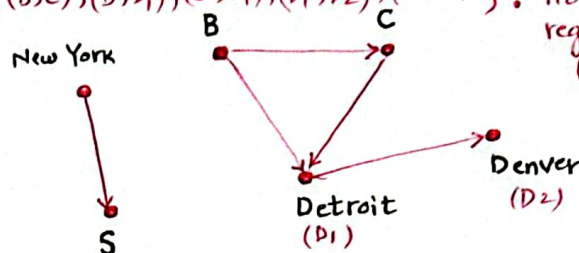


9.4 - CLOSURES OF RELATIONS

A computer network has data ~~sea~~ centers in Boston, Chicago, Denver, Detroit, New York, and San Diego. There are direct, one way telephone lines from

- Boston to Chicago
- Boston to Detroit
- Chicago to Detroit
- Detroit to Denver
- New York to San Diego

$R = \{(B,C), (B,D_1), (C,D_1), (D_1,D_2), (N,S)\}$: no information regarding (B, D_2)



Let R be the relation containing (a,b) if there is telephone line from the data center in a to that in b .

How can we determine if there is some (possibly indirect) link composed of one or more telephone lines from one centre to another?

Because not all links are direct, such as the link from Boston to Denver that goes through Detroit, R cannot be used directly to answer this. Here R is :

$R = \{(B,C), (B,D_1), (C,D_1), (D_1,D_2), (N,S)\}$: no information about the link of Boston and Denver

i.e R is not transitive relation, i.e., it does not contain all the pairs that can be linked.

We can find all pairs of data centres that have a link by constructing a transitive relation R_{new} containing R such that R_{new} is a subset of every transitive relation containing R .

Here, R_{new} is the smallest transitive relation that contains R .

this relation is called the **TRANSITIVE CLOSURE** of R .

In general, Let R be the relation on a set A ,

R may or may ^{not} have some property P such as reflexivity, symmetry or transitivity. If there is a relation R_{new} with property P containing R such that R_{new} is a subset of every relation with property P containing R , then R_{new} is called the **closure** of R with respect to P .

NOTE:
closure of a relation with respect to a property may not exist.

REFLEXIVE CLOSURE

2

Consider the relation

$$R = \{(1,1), (1,2), (2,1), (3,2)\}$$

on the set $A = \{1,2,3,4\}$.

The above relation is not reflexive as $(1,1)$ is there but $(2,2), (3,3), (4,4)$ is not there (Reflexive: $\forall a (a,a) \in R$).

How can we produce a reflexive relation, i.e. R_{new} containing R is as small as possible?

REFLEXIVE CLOSURE of R on a set A is the smallest reflexive relation that contains R and is denoted by R_r^+ (or R_{new}). This relation is a subset of every reflexive relation that contains R .

Given a relation R on a set A , the reflexive closure of R can be formed by adding to R all pairs of the form (a,a) with $a \in A$, not already in R .

$$\begin{aligned} R_r^+ &= R \cup \Delta \quad \text{where } \Delta = \{(a,a) \mid a \in A\} \quad A = \{1,2,3,4\} \\ &= \{(1,1), (1,2), (2,1), (3,2)\} \cup \{(a,a) \mid a \in A\} \\ &= \{(1,1), (1,2), (2,1), (3,2), (2,2), (3,3), (4,4)\} \end{aligned}$$

This new relation is reflexive and is the smallest relation containing R .

This new relation is

- reflexive
- containing R
- the smallest
- subset of every reflexive relation containing R .

Example :

What is the reflexive closure of the relation $R = \{(a,b) \mid a < b\}$ on the set of integers?

Sol:

The reflexive closure of R is

$$\begin{aligned} R \cup \Delta &= \{(a,b) \mid a < b\} \cup \{(a,a) \mid a \in \mathbb{Z}\} \\ &= \{(a,b) \mid a \leq b\} \end{aligned}$$

NOTE: We add only those terms which are needed to make the relation reflexive.

SYMMETRIC CLOSURE:

3

Consider the relation on set $A = \{1, 2, 3\}$ and is not symmetric.

$$R = \{(1,1), (1,2), (2,2), (2,3), (3,1), (3,2)\}$$

: not symmetric as
(2,1) and (1,3)
are not there

How can we produce a symmetric relation containing R is as small as possible?

Symmetric Closure of R on set A is the smallest symmetric relation that contains R . This relation is a subset of every symmetric relation that contains R .

Symmetric ~~Relation~~ Closure of R can be formed by adding all ordered pairs of the form (b,a) (where $(a,b) \in R$) that are not already in the R .

$$= R \cup R^{-1} = R \cup \{(b,a) \mid (a,b) \in R\} \quad \text{: taking union with the inverse}$$

$$\begin{aligned} &= \{(1,1), (1,2), (2,2), (2,3), (3,1), (3,2)\} \cup \{(2,1), (1,3)\} \\ &= \{(1,1), (1,2), (2,2), (2,3), (3,1), (3,2), (2,1), (1,3)\} \end{aligned}$$

Adding these pairs produces a relation that is symmetric, that contains R , that is contained in any symmetric relation that contains R .

Example:

What is the symmetric closure of the relation $R = \{(a,b) \mid a > b\}$ on the set of positive integers?

Sol:

$$R = \{(a,b) \mid a > b\} \quad \text{: not symmetric}$$

Symmetric Closure of R :

$$\begin{aligned} R_{\text{new}} &= R \cup R^{-1} \\ &= \{(a,b) \mid a > b\} \cup \{(b,a) \mid a > b\} \\ &= \{(a,b) \mid a \neq b\} \end{aligned}$$

first element is greater than the second
1st element is less than the second

TRANSITIVE CLOSURE

Consider the relation on the set $A = \{1, 2, 3, 4\}$ and is not transitive.

$$R = \{(1, 3), (1, 4), (2, 1), (3, 2)\}$$

This relation is not transitive because it does not contain all pairs of the form (a, c) where (a, b) and (b, c) are in R .

Can the transitive closure of a relation R be produced by adding all the pairs of the (a, c) , where $(a, b), (b, c) \in R$?

The answer is No, as:

$$\begin{aligned}
 R &= \{(1, 3), (1, 4), (2, 1), (3, 2)\} \\
 &= R \cup \{(a, c) \mid (a, b) \in R, (b, c) \in R\} \\
 R_{\text{new}} &= \{(1, 3), (1, 4), (2, 1), (3, 2)\} \cup \{(1, 2), (2, 3), (2, 4), (3, 1)\} \\
 R_{\text{new}} &= \{(1, 3), (1, 4), (2, 1), (3, 2), (1, 2), (2, 3), (2, 4), (3, 1)\}
 \end{aligned}$$

All ordered pairs of the form (a, c) if $(a, b) \in R$ and $(b, c) \in R$

old member This new relation is transitive because it does not contain $(3, 4)$ as $(3, 2)$ and $(2, 4)$ are in the new relation.

new member $(2, 4)$

This shows that constructing the transitive closure is more complicated than constructing either the reflexive or symmetric closure.

The transitive closure of a relation can be found by adding new ordered pairs that must be present and then repeating this process until no new ordered pairs are needed.

Here we ^{will} discuss two algorithms for constructing transitive closure.

TRANSITIVE CLOSURE:

Finding the transitive closure of a relation is equivalent to determining which pairs of vertices in the associated directed graph are connected by a path.

With this in mind, we define a new relation

DEF.

CONNECTIVITY RELATION:

Let R be a relation on a set A . The Connectivity relation R^* consists of the pair (a, b) such that there is a path of length at least one from a to b in R .

$$R^* = \bigcup_{n=1}^{\infty} R^n \quad : \quad R^n \text{ consists of the pairs } (a, b) \text{ such that there is a path of length } n \text{ from } a \text{ to } b.$$

Theorem:

The transitive closure of a relation equals the connectivity relation R^* .

Lemma:

Let A be a set with n elements, and Let R be a relation on A .

If there is a path of length at least one in R from a to b , then there is such a path with length not exceeding n .

Moreover, when $a \neq b$, if there is a path of length at least one in R from a to b , then there is such a path with length not exceeding $n-1$.

From Lemma, we see that

the transitive closure of R is the union of $R, R^2, R^3, \dots, R^n$.

$$R^* = R \cup R^2 \cup R^3 \cup \dots \cup R^n$$

Theorem:

Let M_R be the zero-one matrix of the relation R on a set with n elements. Then the zero-one matrix of the transitive closure R^* is

$$M_{R^*} = M_R \vee M_R^{[2]} \vee M_R^{[3]} \vee \dots \vee M_R^{[n]}$$

ALGORITHM: Computing Transitive Closure
transitive closure (M_R : zero-on $n \times n$ matrix)

$A := M_R$

$B := A$

for $i = 2$ to n

begin

$A := A \odot M_R$

$B := B \vee A$

end { B is the zero-one matrix for R^* }

EXAMPLE:

Find the zero-one matrix of the transitive closure of the relation R , where

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

Sol:

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$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$M_R^{(2)} = M_R \odot M_R$$

$$M_R^{(2)} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \odot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 \vee 0 \vee 0 & 0 \vee 0 \vee 1 & 1 \vee 0 \vee 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$M_R^{(3)} = M_R^{(2)} \odot M_R$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \odot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

To find this matrix, the successive Boolean powers of M_R , upto the n th power, are computed.

As each power is calculated, its join with the join of all small powers is formed.

Now

$$M_R^* = M_R \vee M_R^{(2)} \vee M_R^{(3)}$$

$$= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\text{Time Complexity} = O(n^4)$$

$$\text{Bit operations} = n^2(2n-1)(n-1) + (n-1)n^2$$

$$= 2n^3(n-1)$$

$$= 2n^4 - 2n^3$$

$$= O(n^4)$$

The remainder of this section describes more efficient algorithm.

Warshall's algorithm is an efficient method for computing the transitive closure of a relation. Time Complexity of warshall's algorithm is $2n^3$ bit operations.

$$O(n^3)$$

Example:

Use Marshall's Algorithm to find Transitive ~~Algorithm~~ Closure of R.

Sol,

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \quad A = \{1, 2, 3\}$$

Step-I:

Take

$$W_0 = M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\text{1st Col: } C_1 = \{1, 3\} \quad \text{1st row: } R_1 = \{1, 3\}$$

$$C_1 \times R_1 = \{(1, 1), (1, 3), (3, 1), (3, 3)\}$$

$$W_1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Step-II:

$$\text{2nd Col: } C_2 = \{2, 3\}$$

$$\text{2nd row: } R_2 = \{2\}$$

$$C_2 \times R_2 = \{(2, 2), (3, 2)\}$$

$$W_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Step-III:

$$\text{3rd Col: } C_3 = \{1, 3\}$$

$$\text{3rd row: } R_3 = \{1, 2, 3\}$$

$$C_3 \times R_3 = \{(1, 1), (1, 2), (1, 3), (3, 1), (3, 2), (3, 3)\}$$

$$W_3 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Hence, $R^* = W_3$ is the transitive closure of R.

$$R^* = \{(1, 1), (1, 2), (1, 3), (2, 2), (3, 1), (3, 2), (3, 3)\}$$

$$W = M_R$$

for $k=1$ to n

for $i=1$ to n

for $j=1$ to n

$$w_{ij} = w_{ij} \vee (w_{ik} \wedge w_{kj})$$

$$\{W = [w_{ij}] \text{ is } M_{R^*}\}$$